This is the author's version of an article that has been published in this journal. Changes were made to this version by the publisher prior to publication.
The final version of record is available at http://dx.doi.org/10.1109/TCBB.2019.2941689

On Axon Interaction and Its Role in Neurological Networks

Aman Chawla, *Member, IEEE,* Salvatore Morgera, *Fellow, IEEE,* and Arthur Snider

Abstract—The study of axon tracts is important in neurophysiology. In this paper, we study axon tracts with arbitrary geometry and present a governing equation for ephaptic coupling among the constituent axons. We derive this equation and simulate it, obtaining a novel, geometry-dependent, delayed coupling phenomenon. This has relevance for how an axon tract communicates data in a neurological network.

Index Terms—ephaptic coupling, tract geometry, myelinated nerve fibers, action potential, synchronization.

1 Motivation and Review: Why do we need realistic models of axon tracts?

1.1 Motivation

Our study is motivated by the need to deepen understanding of the nervous system, in general, and neurological disorders, such as multiple sclerosis and autism spectrum disorder, in particular. In this paper we focus on the phenomenon of ephaptic coupling. This phenomenon is chosen since it could have relevance for a number of neurological disorders.

The nervous system can be seen as a communication network. We ultimately seek to understand how critical elements of the nervous system act as network elements to process information. For this, a thorough understanding of the physical layer[1] of the nervous system network is essential. Within the physical layer we focus on a very specific phenomenon - the phenomenon of ephaptic interaction and the concomitant synchronization.

Multiple sclerosis is a significantly prevalent disorder of the nervous system with no known cure. While there is some understanding of the mechanism underlying the disease, the exact impact of demyelination on the operation of the nervous system is not known. There have been several studies on modeling nervous system conduction under diseased conditions [1], [2], but no known study has incorporated the geometrical arrangement of axons in a tract into the model and simulations, which we do here. This is important since multiple sclerosis leads to demyelination plaques in areas such as the brain, optic nerve and spinal cord, where nerve fibers are tightly bundled in tracts. Here the packing is not necessarily homogeneous or uniform, and we cannot simply reduce the intricacy of the system to a single dimension. Our simulations reveal that angular inclination and relative placement of axons does make a difference in the abilities of a tract for conducting coupled action potentials.

1.2 Historical Review

The pioneering work by Hodgkin and Huxley on the squid giant axon set the stage for all future developments in the field of axon conduction. In 1942, Arvanitaki [3] proposed the use of the term "ephapse" and, consequently, ephaptic. His work contains the experimental results of studying the interaction between two axons placed side by side. This seminal work introduces some of the key concepts such as looking at the electric potential gradients in the zone of interaction between the two ephaptically coupled axons, which to this date has not been simulated.

According to Markin [4], who focuses on resistive potential drops as mediating the interaction between coupled axons, the main difficulty in producing an analytical solution to the problem of interaction between two fibers is the intractability of analytically solving the Hodgkin Huxley [5] equations for the ionic currents. He gets around this problem by proposing a simplifying model for the ionic current. He then uses this model to derive a set of governing equations based on a simple resistive-capacitive (R-C) network representing two interacting fibers, interacting by means of ion current flow in and out of the shared extracellular space.

Markin then goes on to examine the solutions to this model and finds, in the case when there is a spike on only the first fiber, that there are two possible values of impulse velocity, one stable and the other unstable, with stability being defined in relation to persistence of the impulse at that velocity. He also examines the case in which a spike on the first fiber excites a spike on the second one. This case occurs only when the second fiber has a diminished threshold for action potential initiation. In companion papers [6], [7] Markin considers the collective conduction of spikes as well as their interaction in bundles of fibers.

The same problem of the analytical intractability of the Hodgkin Huxley equations is encountered by Eilbeck, Scott and Luzader [8]. They address the problem by using the FitzHugh-Nagumo set of equations instead of the Hodgkin Huxley equations and solving that system by a perturbation analysis. They also perform simulations on the coupling of identical fibers. This work is primarily analytical, focusing only on two fibers. Furthermore, they do not consider the geometrical arrangement of many fibers in a tract.

1. The layer which consists of the physical medium or channel of communication

Symbol	Meaning
C	myelin capacitance per unit length
$G^{ax,my,ext}$	conductance per unit length (axoplasm, myelin, extracellular)
$V_{i,p}$	Transmembrane potential drop (axon i or axon p)
θ_i	Inclination of axon i with respect to tract axis
s_i	Spatial position along an inclined axon
$r_{inner,outer}$	(inner, outer) myelin sheath radius
z	Spatial position along the tract axis
$\mathbf{r}$	Position vector of a point in the tract
$\phi(\mathbf{r})$	Electric potential at a point $\mathbf{r}$ of the tract
$\phi_{ext}(z,t)$	Extracellular potential at a spatial height z along the tract axis
$\phi_i(s_i,t)$	Intracellular potential of axon i at a spatial distance s_i along the axon
$\mathbf{r}_i$	Location of the base of axon i
$\mathbf{n}_i$	Unit vector along axon i
I	current (axoplasm, myelin, extracellular, injected or ionic)
A	cross-sectional area (axoplasm, myelin, extracellular)
$\sigma_{ax,my,ext}$	conductivity (axoplasm, myelin, extracellular)
ϵ_{my}	permittivity of myelin
W_{ip}	phenomenological distance factor between axons i and p
N	number of axons in a tract

TABLE 1
List of principal symbols and their meanings

Reutskiy, Rossoni and Tirozzi [1] modify the Frankenhaeuser Huxley [9] set of equations and extend them to incorporate ephaptic coupling via an assumption on the electrical isolation of the tract, which is also present in Markin's papers. They do not provide an analytical solution, but present the results of computer simulations. Their work, which provides a fiducial point for our present paper, incorporates large sized tracts with many fibers, all identical, with no reference to their geometrical arrangement.

In the present decade, one of the most comprehensive reviews of ephaptic coupling is the paper by Anastassiou and Koch [10]. They conjecture that ephaptic coupling might rely on a feedback mechanism between the extracellular electric field and the intracellular activity that generates it in the first place. Providing further motivation for our work, they emphatically state that

> "How … ephaptic effects are mediated and manifested in the brain remains a mystery. … argue that it is both possible and worthwhile to vigorously pursue such research as it has significant implications on … understanding of brain processing …."

Their paper includes a comprehensive survey of the literature and comprises both experimental and computational work. We refer the reader to the references in this work for further background on ephaptic coupling [2].

2. As a note to the reader, while this paper will focus on current-mediated coupling between axons, there is also a possibility for field-mediated coupling [[11], Chapter 4] which finds a growing body of experimental support [12], [13]. These two mechanisms are not entirely unrelated [[11], Chapter 6].

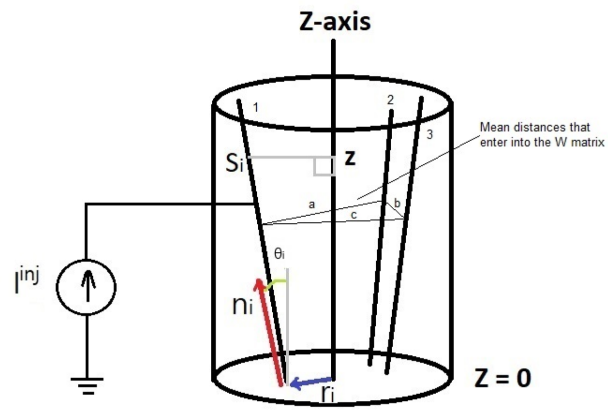


Fig. 1. A cylindrical axon tract aligned with the z-axis is shown, with three axons, 1, 2, and 3, inclined with respect to the z-axis, shown within it. The relationship $z = s_i \cos\theta_i$ holds between the variable along the axon and the variable along the axis. The representative cross-sectional inter-axonal distances between 1 and 2, 2 and 3, and 3 and 4 are labeled as a, b and c respectively.

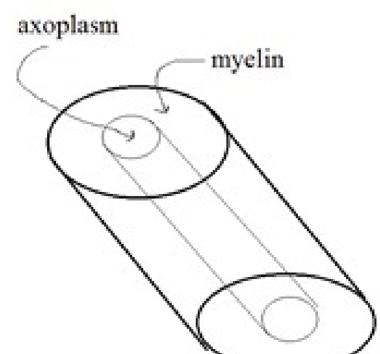


Fig. 2. A single, tilted, cylindrical axon is shown with myelin sheath and enclosed axoplasm. In cross-section, the inner area is πr_{inner}^2 while the sheath area is $\pi(r_{outer}^2 - r_{inner}^2)$

2 Derivation of Generalized Model for Interaction between Axons

Figure 1 specifies the geometric parameters for a tract of axons, encased in a right cylinder aligned with the z axis. The ith axon itself consists of sections of cylinders of axoplasm, surrounded by myelin sleeves, connecting the nodes of Ranvier (Figure 2). The axons can have various cross sectional areas, material parameters, and misalignments with the tract axis.

Note that each point on the ith axon has a local coordinate given by its distance s_i along the axon, and a universal coordinate $z = s_i \cos\theta_i$. Derivatives are then related by the chain rule

$$\frac{\partial}{\partial s_i} = \cos\theta_i \frac{\partial}{\partial z}.$$

The electric potential throughout the tract is denoted $\phi(\mathbf{r})$. Our model invokes simplifying assumptions that spawn two different regional formulations for ϕ.

1) As per the authors of [[1], p.3], each transverse section of the extracellular space is isopotential; that

is, in the extracellular space $\phi(\mathbf{r}, t)$ is a function only of z:

$$\phi(\mathbf{r}, t) = \phi_{ext}(z, t).$$

Nevertheless, $\phi_{ext}(z, t)$ depends (see (8)) on the electrical effects produced inside all the axons. It is in fact the mechanism for the ephaptic coupling between them.

2) Inside the ith axon fiber, $\phi(\mathbf{r}, t)$ varies only along the fiber length:

$$\phi(\mathbf{r}, t) = \phi_i(s_i, t).$$

Here $\mathbf{r} = \mathbf{r}_i + s_i\mathbf{n}_i$.

Ion currents obeying the Frankenhauser-Huxley equations are generated at the nodes of Ranvier, and the voltage propagation along the fiber is modeled by the well known cable equation [5], [1], [14]. It is derived by observing that the change in (longitudinal) current I, over a differential length ds, must be accounted for by incoming transversal currents through the myelin sheath (capacitive, ohmic and ionic or injected):

$$I(s + ds) - I(s) = -C ds \frac{\partial V}{\partial t} - G^{my} ds V + I^{inj} ds \quad (1)$$

or,

$$\frac{\partial I}{\partial s} = -C\frac{\partial V}{\partial t} - G^{my}V + I^{inj}, \quad (2)$$

where $V(s, t) \equiv \phi(s, t) - \phi_{ext}(z \equiv s\cos\theta, t)$ is the voltage drop across the myelin sheath, $C = \frac{2\pi\epsilon_{my}}{\ln\frac{r_{outer}}{r_{inner}}}$, $G^{my} = \frac{2\pi\sigma_{my}}{\ln\frac{r_{outer}}{r_{inner}}}$, and $I^{inj}(s, t)$ denotes any injected current. We ignore longitudinal myelin currents. For clarity we have temporarily suppressed the subscripts that associate the physical and geometric parameters with particular axons. In all that follows the Frankenhauser-Huxley nodal current is also accounted for by the placeholder I^{inj}.

Since I is driven by the gradient of the potential, $I = -G^{ax}\frac{\partial\phi}{\partial s}$, ($G^{ax} = \sigma_{ax}\pi r_{inner}^2$) and the cable equation takes the familiar form

$$\begin{aligned} C\frac{\partial V}{\partial t} &= G^{ax}\frac{\partial^2\phi}{\partial s^2} - G^{my}V + I^{inj} \\ &\equiv G^{ax}\cos^2\theta\frac{\partial^2[V + \phi_{ext}]}{\partial z^2} - G^{my}V + I^{inj}. \end{aligned} \quad (3)$$

The occurrence of the extracellular potential in this equation gives rise to ephaptic coupling. We have a cable equation for each axon voltage V_i; we need an equation for $\phi_{ext}(z, t)$. But, in fact, the extracellular region itself can be modeled as a cable, wherein longitudinal variation in its current $I_{ext}(z, t)$ is accounted for by the total transversal currents from the axons:

$$\begin{aligned} I_{ext}(z + dz) - I_{ext}(z) &= \sum_{axons,p} C_p ds_p \frac{\partial V_p}{\partial t} \\ &+ G_p^{my} ds_p V_p - I_p^{inj} ds_p \\ &\equiv -\sum_{axons,p}\frac{\partial I_p}{\partial s_p} ds_p, \end{aligned} \quad (4)$$

or,

$$\frac{\partial I_{ext}}{\partial z} = -\sum_{axons,p}\frac{\partial I_p}{\partial s_p}\frac{ds_p}{dz} = -\sum_{axons,p}\frac{\partial I_p}{\partial z}. \quad (5)$$

That is,

$$I_{ext} + \sum_{axons,p} I_p = \text{constant}. \quad (6)$$

Here $\sum_{axons,p}$ stands for a summation over the indices of all the axons, that is, p ranges from $1, \ldots, N$ where N is the total number of axons in the tract. Expressed in terms of the potential, (6) says

$$\begin{aligned} -G^{ext}\frac{\partial^2\phi_{ext}}{\partial z^2} &- \sum_{axons,p} G_p^{ax}\frac{\partial}{\partial z}\frac{\partial\phi_p}{\partial s_p} \\ &= 0 \\ &= -G^{ext}\frac{\partial^2\phi_{ext}}{\partial z^2} - \sum_{axons,p} G_p^{ax}\cos\theta_p\frac{\partial^2\phi_p}{\partial z^2} \end{aligned} \quad (7)$$

where $G^{ext} = \sigma_{ext}A_{ext}$. Expressing the axon potentials as $\phi_p = V_p + \phi_{ext}$, we obtain an equation for the extracellular potential

$$\begin{aligned} -G^{ext}\frac{\partial^2\phi_{ext}}{\partial z^2} &- \sum_{axons,p} G_p^{ax}\cos\theta_p\frac{\partial^2\phi_{ext}}{\partial z^2} \\ &- \sum_{axons,p} G_p^{ax}\cos\theta_p\frac{\partial^2 V_p}{\partial z^2} \\ &= 0 \end{aligned}$$

or

$$\frac{\partial^2\phi_{ext}}{\partial z^2} = \frac{-\sum_{axons,p} G_p^{ax}\cos\theta_p\frac{\partial^2 V_p}{\partial z^2}}{\left[G^{ext} + \sum_{axons,p} G_p^{ax}\cos\theta_p\right]}. \quad (8)$$

We substitute for $\frac{\partial^2\phi_{ext}}{\partial z^2}$ in the axon cable equations (3) to derive the generalized ephaptic equation

$$\begin{aligned} C_i\frac{\partial V_i}{\partial t} &= G_i^{ax}\cos^2\theta_i\left(\frac{\partial^2 V_i}{\partial z^2} - \sum_{axons,p} K_p\frac{\partial^2 V_p}{\partial z^2}\right) \\ &- G_i^{my}V_i + I_i^{inj} \end{aligned} \quad (9)$$

where

$$K_p = \frac{G_p^{ax}\cos\theta_p}{G_{ext} + \sum_{axons,p} G_p^{ax}\cos\theta_p}.$$

This equation reduces to that of Reutskiy [1] if the misalignment effects (θ_i) are ignored and the axons are all identical. It further reduces to that of Hodgkin and Huxley [5] for the case of a single axon.

The authors feel it is important to note that some workers have been troubled by Reutskiy's postulate that the net longitudinal current flux across any transversal plane must be *zero*: $I_{ext}(z) + \sum_{axons,p} I_p(z) = 0$. In fact, the postulate is false. One can impose a uniform longitudinal electric field E along the tract, decreasing every potential by a term proportional to z ($\phi_p' = \phi_p - Ez$, $\phi_{ext}' = \phi_{ext} - Ez$); the cable equations (3) and (7) will still be satisfied, and there will be a nonzero current flux (independent of z). Reutskiy, however, only employed the *derivative* of this condition (5) in his derivation, so it is consistent with the present analysis.

The term $K_p \frac{\partial^2 V_p}{\partial z^2}$ quantifies how the contribution to the extracellular potential by the pth axon influences the voltage in the ith axon. Intuitively, we feel that a more sophisticated model of the extracellular potential (without the simplifying first assumption of no transversal variation of the extracellular potential) would predict an attenuation of this effect as the distance between the two axons increases. In the absence of such a model, we phenomenologically propose that a factor W_{ip} be inserted into the equation (9), diminishing with separation between axons i and p:

$$\begin{aligned} C_i \frac{\partial V_i}{\partial t} &= G_i^{ax} \cos^2 \theta_i \left[\frac{\partial^2 V_i}{\partial z^2} - \sum_{axons,p} W_{ip} K_p \frac{\partial^2 V_p}{\partial z^2} \right] \quad (10) \\ &- G_i^{my} V_i + I_i^{inj} \end{aligned}$$

where i runs from $1, \ldots, N$. The following section reports simulations that were performed incorporating the various innovations described.

3 Simulation

3.1 Tract injury: Generalized ephaptic equation simulation

To illustrate the power and importance of our formulation, in this subsection we consider the case of a compressive injury (lateral compression in a cross-sectional plane) to a tract containing three parallel axons resulting in a change in the angular inclinations of the axons to the tract axis, by thirty degrees each, but no change in the mean separation between the axons (i.e. no change in the W_{ip} matrix). The axons are placed in a non-equilateral arrangement with the W_{ip} matrix entries being:

$$\begin{bmatrix} 1 & 0.892 & 0.99 \\ 0.892 & 1 & 0.9001 \\ 0.99 & 0.9001 & 1 \end{bmatrix} \quad (11)$$

The matrix is illustrated schematicallly in Fig. 1 with the three representative dimensions being marked as a, b and c. The values in the above matrix were chosen arbitrarily, but satisfying the condition that the sum of two sides of a triangle must be larger or equal to the third side. We also kept in mind an inverse relation between an entry and the corresponding distance. We simulate the phenomenological version of the ephaptic equation (10) for this case before the injury with $\theta_i = 0$. This simulation result primary case is shown in Fig. 3. The secondary cases are summarized in the following by providing only the action potential initiation times on the various axons. In the first secondary case, axon two is stimulated. The initiation time of spikes on axon one is 2.35 ms and on axon three is 2.2 ms. In the second secondary case, axon three is stimulated. The initiation time of spikes on axon one is 1.35 ms, while on axon two is 1.6 ms.

Next, for the post-injury case, we simulate the same equation for the case of three symmetrically inclined axons ($\theta_i = 30$) and W_{ip} being unchanged. The primary simulation result is shown in Fig. 4. The secondary simulation results are indicated in the following. In the first secondary case, axon two is stimulated. Axons one and three show only subthreshold activity. In the second secondary case axon three is stimulated. The spike initiation time on axon one is 2.3 ms while axon two shows only subthreshold activity.

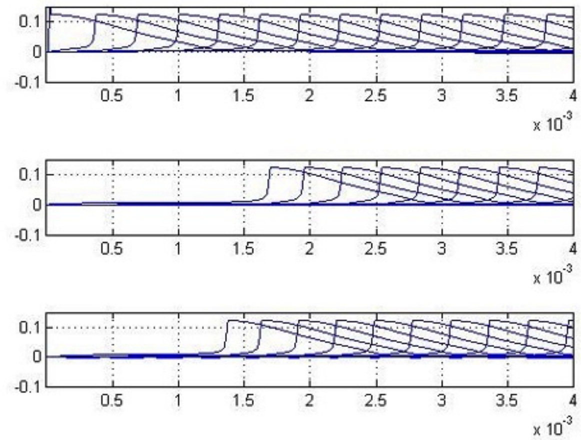


Fig. 3. Three axons of length 6 cm are considered (axon 1, top; axon 2, middle; axon 3, bottom) with a W_{ip} which incorporates non trivial inter-axonal distances and inclinations of $\theta_i = 0$ from the tract axis and a tract diameter of approximately 2.71 mm. Axon 1 is stimulated. The x-axis is time in seconds and the y-axis is voltage in volts.

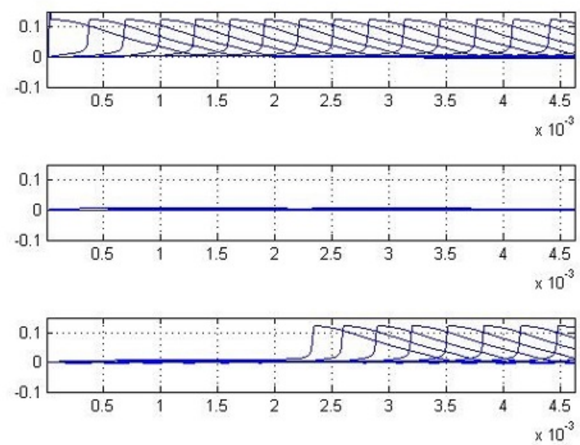


Fig. 4. Three axons are considered (axon 1, top; axon 2, middle; axon 3, bottom) with a W_{ip} which incorporates non trivial inter-axonal distances and inclinations of $\theta_i = 30$ from the tract axis. Axon 1 is stimulated. The x-axis is time in seconds and the y-axis is voltage in volts.

From these six cases, we can infer that the injury has a dramatic impact on the ephaptic coupling. Before injury, when axon one is stimulated, axons two and three both fire (Fig. 3). However, after injury, only axon three fires in response (Fig. 4). Similarly, when axon two is stimulated before injury, axons one and three would also fire, but after injury they do not respond. Likewise, when axon three is stimulated, before injury axons one and two both fire, whereas after injury axon two does not respond.

Thus we see that the injury has tended to cut-off axon two's communication with the other two axons, either wholly or by introducing severe delay. This changes not

This is the author's version of an article that has been published in this journal. Changes were made to this version by the publisher prior to publication.
The final version of record is available at http://dx.doi.org/10.1109/TCBB.2019.2941689

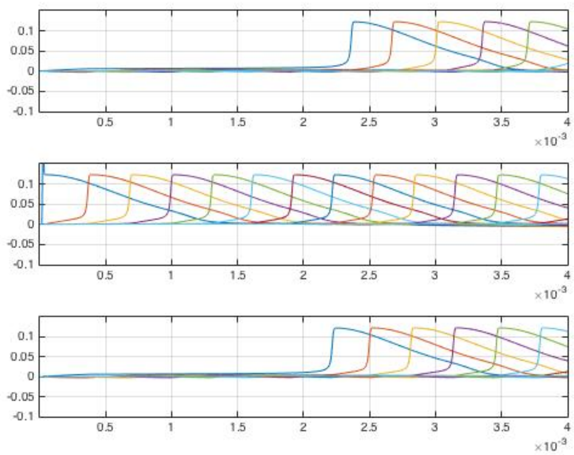


Fig. 5. Three axons are considered (axon 1, top; axon 2, middle; axon 3, bottom) with a $W_{12} = 0.892$ and the other distances given in (11). Axon 2 is stimulated. The x-axis is time in seconds and the y-axis is voltage in volts.

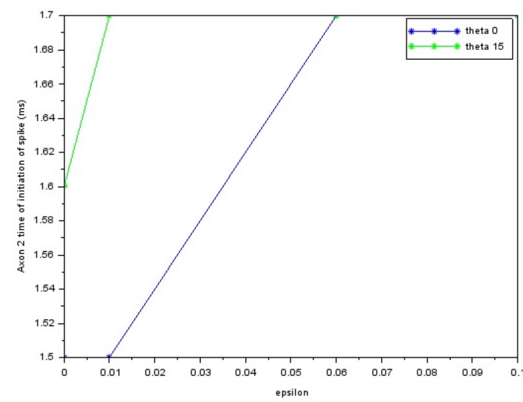


Fig. 6. Axon 2 spike initiation times (in ms) with fixed inclination, varying ϵ; for different θ (zero degrees - blue and fifteen degrees - green). The θ of 30 degrees result is shown in Fig. 4. It is apparent that the slope of the green line is greater than that of the blue line. As θ increases the slope is increasing until it becomes vertical for theta of 30 degrees case - the corresponding axon is put out of communication, either wholly or due to very large delay.

only the configuration but also the functionality of the axon interaction network.

3.2 Gradual variation of interaxonal distances: Simulation

In another example, we systematically vary one of the three interaxonal distances in the W_{ip} matrix, to see its effect on the timing of coupled action potential onset. Specifically, axon two is stimulated. We vary gradually the distance between axon two and axon one and plot the time sequence of action potentials of all three axons, as this distance is increased. The other distances are the same as in (11). Fig. 5 shows the output for the parameter W_{12} being set to 0.8920. The cases of W_{12} being 0.8539 and 0.7570 respectively are summarized descriptively in the following. When $W_{12} = 0.8539$ and axon two is stimulated, axon one initiation time is 3.1 ms and axon three spike initiation time is 2.5 ms. When $W_{12} = 0.7570$ and axon two is stimulated, axons one and three show only subthreshold activity.

3.3 Further investigations of impact of joint variation of W_{ip} and θ

We parametrized the W_{ip} matrix by a parameter ϵ which was added to one arm of the triangle and subtracted from another arm, such that the three arms always satisfied the relation that the sum of any two sides of a triangle is greater than the third side.[3] As we varied ϵ and the common inclination of the axons, θ, we measured axon two's time of spike initiation when axon one was stimulated and axon three's time of spike initiation similarly. The first case we studied was for fixed inclination, but varying ϵ; for different values of inclination θ. The second case we studied was for fixed ϵ, but varying θ; for different values of ϵ.

We found in the first case (Figures 6 and 7 below) a nonlinear relation between ϵ and the time of action potential initiation in both the unstimulated axons for different

3. Note that ϵ used here is distinct from ϵ_{my} appearing in Section 2

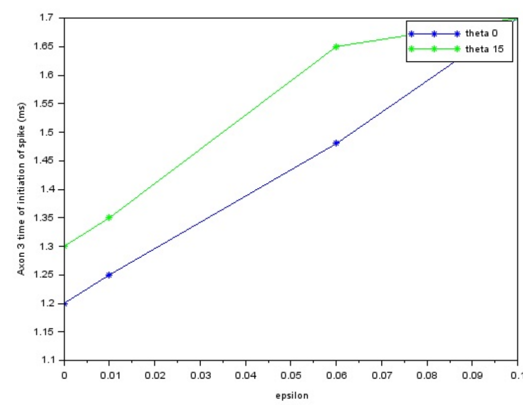


Fig. 7. Axon 3 spike initiation times (in ms) with fixed inclination, varying ϵ; for different values of θ (zero degrees - blue and fifteen degrees - green)

inclinations. In the second case also there was a nonlinear relation between the inclination θ and the time of spike initiation on both the axons for different values of ϵ (Figures 8 and 9 below). We further noted that when the W_{ip} matrix is varied slightly, there is only a slight perturbation in this nonlinear relationship, indicating a possible differentiability of the initiation-time vs. θ curve with respect to the third degree of freedom, ϵ. The case beyond 35 degrees of inclination was not investigated as it would require greater simulation time to obtain supra-threshold responses on the unstimulated axons.

3.4 Synchronization: Simulation of differing diametered axons

Although synchronization can be studied for the case of three axons, we come back to a simpler case, that of two parallel axons and in this subsection study *synchronization* between the impulses launched on them. Since there are

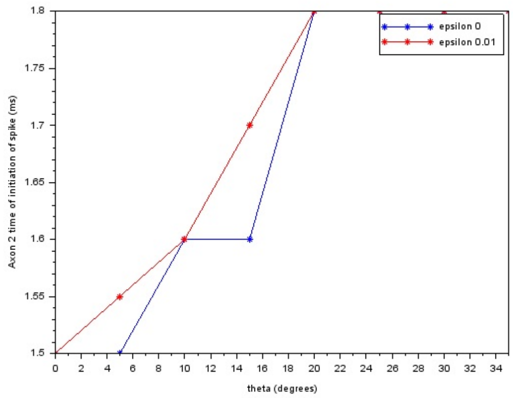


Fig. 8. Variation of axon 2 spike initiation time (in ms) with θ; for different values of ϵ (zero - blue and 0.01 - red)

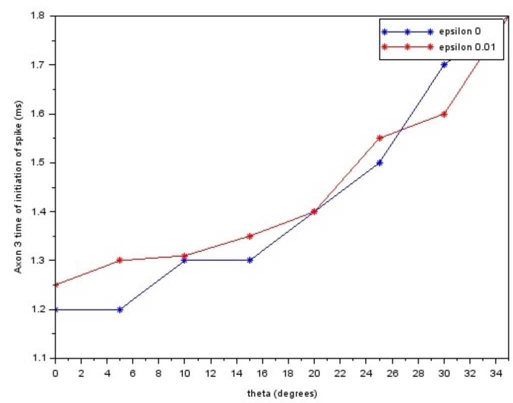


Fig. 9. Variation of axon 3 initiation time (in ms) with θ; for different values of ϵ (zero - blue and 0.01 - red)

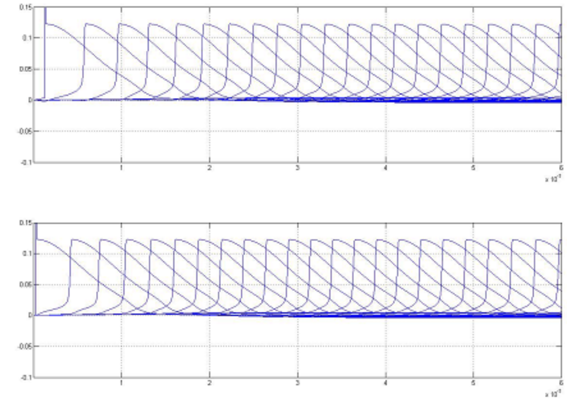


Fig. 10. Two axons are considered, the second with 10 percent larger diameter than the first one (of diameter 10 microns and length 8 cm). A smaller value of extracellular resistance is used to show the coupled and synchronizing propagation on the two fibers. The second one here (bottom) is stimulated before the first one.

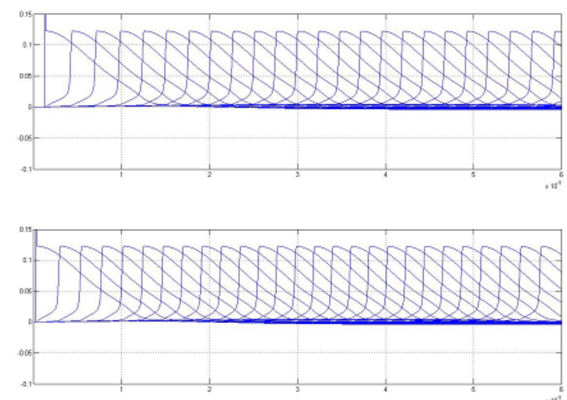


Fig. 11. Two axons are considered, the second with 10 percent larger diameter than the first one. A null value of extracellular resistance is used to show the propagation under lack of coupling for the sake of comparison with the previous figure. The second axon here (bottom) is stimulated before the first one.

only two parallel axons, the full power of the generalized ephaptic equation is not needed and we can simply use the special case without W_{ip} and with $\theta_i = 0$.

The case of synchronization between differing diametered axons has not been simulated to our knowledge. The analysis presented in the book [15] reviews how the velocities on fibers with different diameters synchronize with time. That analysis is summarized in Appendix C. This presentation forms the motivation to simulate the case where the fibers being coupled have different diameters, for different diameters lead to different conduction velocities. We note that a tract may contain a range of axonal diameters. See, for example, Fig. 7 in [16].

To study the impact of different diameters, and eventually obtain Figs. 12 and 13, we computed C, G^{ax} and G^{my} for different-sized axons. The results obtained for the case of two different diameters are shown in Figs. 10 and 11. The figures clearly demonstrate that the action potentials on the unstimulated fiber which are initially lagging behind those on the stimulated fiber, speed up to catch up with the former. These simulation results agree with the analysis in [15] which indicates that the lagging pulse speeds up to catch up to the leading pulse. A more extensive set of simulation runs would be required to completely characterize what happens in different situations when fiber diameters differ. Because this is a nonlinear phenomenon, it might not be possible to exhaust all the possibilities by simulation runs. However, the following two figures, Fig. 12 and 13, are a fair way to capture the essence of what is happening in synchronization. They show the spatial sum of the action potentials on two axons, standing alone (Fig. 13) and when coupled and synchronizing (Fig. 12). A beat-like phenomenon and eventual synchronization is clearly visible in Fig. 12.

4 Limitations, Conclusions and Future Work Directions

In this work, we developed a new method for incorporating the geometrical arrangement of axons in a tract into the known cable equation models. This led to a generalized equation of which the known equations such as the

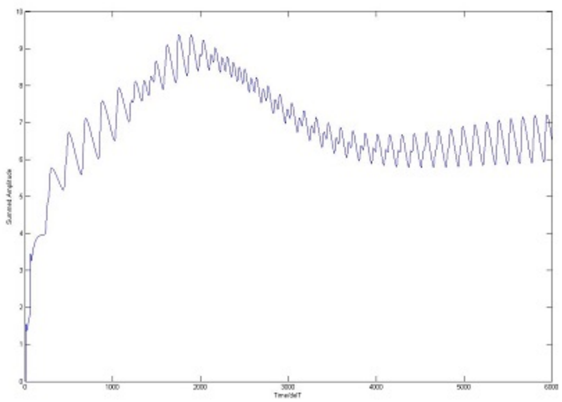


Fig. 12. Signatures of neuronal harmony. Beats and phase synchronization when action potentials on two axons with differing diameters synchronize. To obtain this curve, we simulated the case $N = 2$, with coupling, between axons of two different diameters, differing by 10 percent. The voltage profiles at each node were summed with each other to obtain the curve. On the curve, the left side shows the onset of the 'beats' envelope. The right end shows the eventual perfect phase synchronization.

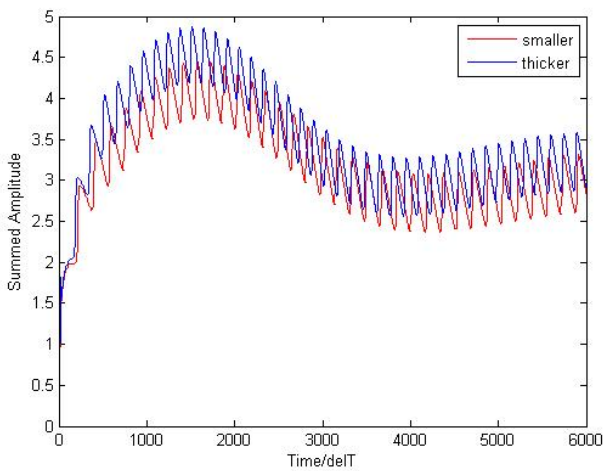


Fig. 13. To obtain these two curves, we simulated twice. Both simulations were for the case $N = 1$, but the axon diameters were varied, one being 10 percent larger than the other. In each simulation, the voltage profiles of all the nodes were summed. This yielded the two curves.

Hodgkin-Huxley equation and the Reutskiy equation are special cases. The simulations of this equation showed that a compressive injury can cut-off an axon from axon-axon interaction. Further, variation of an inter-axonal distance via the parameter matrix W_{ip} has an impact on sympathetic impulse stimulation time. The simulations thus demonstrate the relevance of an equation such as the one derived here, for further investigation of the phenomena involved in ephaptic coupling. Some of the limitations of our work include the homogeneous placement of nodes of Ranvier on each of the axons. This can easily be addressed in future simulations, and is not a limitation of the theoretical development. Further, we did not address tortuous or skewed tracts and axons. This might require differential geometry, which is beyond the scope of this work. Our equations are thus limited to a right circular cylindrical tract of axons.

In a realistic tract of axons, one can determine, by clinical measurements, the precise interaxonal distances at a plane of cross-section along the tract axis. As an example, the rodent cortico-spinal tract has been measured clinically. It has been found that an estimated 15 to 40 (average 25) axons of less than one micron in diameter appear in a cross-sectional area of 25 square microns [17]. Since our work would allow the incorporation of these inter-axonal distances, this highlights the importance of our work to clinicians, for modeling and simulating such a tract. We would like to note here that we tested only the case when all the axons have equal angular inclinations to the tract axis; there is also the case of different angular inclinations. In both cases, there will be small variations with z which we neglect, assuming the inclinations to be small enough over a tract length that a "mean" value is representative.

The work can be extended in a few directions. First of all, we should be able to study multiple sclerosis directly, by incorporating demyelination, specifically focal demyelination, into our program. Focal demyelination is the condition wherein the region around a node gets progressively demyelinated. This might be incorporated by replacing the internodal segments around a node by node-like segments in the program we developed to simulate the generalized ephaptic equation. Further, here we could simulate the variation of capacitance and conductance with distance by introducting a spatial dependence on these variables.

Secondly, we can consider studying the spread of local ephaptic rhythms into global ones via simulation. This can be done by taking a tract with a large number N of axons, say $N = 1000$, and stimulating only a few of them, close to the axis of the tract. The goal then would be to study how fast the ephaptic effect leads to a spread of this stimulation. That is, how soon do the neighbors of the stimulated set get coupled in and begin firing their own action potentials. Another question that can be addressed is whether the whole tract eventually begins to fire or not, and what are the threshold conditions and geometric constraints for this happening.

Finally, we can study the impact of noisy current injections on the input side to a tract, by randomly stimulating the input side. This would constitute an interesting study of academic interest and not just possible clinical impact. Preliminary investigations indicate that noise may be added and that this is equivalent to noisy current injections. If we next treat the axon tract as a data storage device or channel in time, then this setup likely results in a form of discrete time additive noise channel with perfect feedback. The capacity of a number of such channels is known in the information theory literature and therefore we may be able to compute the storage capacity of the axon tract.

Further, we can incorporate applied electric fields as additional current injections and study their impact on the synchronous firing of axons in the tract. To conclude, the theory and the program that has been built as a part of this investigation have many potential extensions and can be useful to neurophysiologists as well as those studying fundamental communication and storage limits of the nervous system.

References

[1] S Reutskiy, Enrico Rossoni, and Brunello Tirozzi. Conduction in bundles of demyelinated nerve fibers: computer simulation. *Biological Cybernetics*, 89(6):439–448, 2003.

[2] Stephen G Waxman and Michael H Brill. Conduction through demyelinated plaques in multiple sclerosis: computer simulations of facilitation by short internodes. *Journal of Neurology, Neurosurgery & Psychiatry*, 41(5):408–416, 1978.

[3] A Arvanitaki. Effects evoked in an axon by the activity of a contiguous one. *Journal of Neurophysiology*, 5(2):89–108, 1942.

[4] VS Markin. Electrical interaction of parallel non-myelinated nerve fibres. I. Change in excitability of adjacent fibre. *Biophysics-USSR*, 15(1):122, 1970.

[5] Alan L Hodgkin and Andrew F Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500, 1952.

[6] VS Markin. Electrical interaction of parallel non-myelinated nerve fibres. II. Collective conduction of impulses. *Biophysics-USSR*, 15(4):713, 1970.

[7] VS Markin. Electrical interaction of parallel non-myelinated nerve fibres. III. Interaction in bundles. *Biophysics*, 18:324–332, 1973.

[8] JC Eilbeck, SD Luzader, and AC Scott. Pulse evolution on coupled nerve fibres. *Bulletin of Mathematical Biology*, 43(4):389–400, 1981.

[9] B Frankenhaeuser and AF Huxley. The action potential in the myelinated nerve fibre of xenopus laevis as computed on the basis of voltage clamp data. *The Journal of Physiology*, 171(2):302, 1964.

[10] Costas A Anastassiou and Christof Koch. Ephaptic coupling to endogenous electric field activity: why bother? *Current Opinion in Neurobiology*, 31:95–103, 2015.

[11] Aman Chawla. *On Axon-Axon Interaction via Currents and Fields*. PhD thesis, University of South Florida, 2017.

[12] Rengasayee Veeraraghavan, Joyce Lin, James P Keener, Robert Gourdie, and Steven Poelzing. Potassium channels in the cx43 gap junction perinexus modulate ephaptic coupling: an experimental and modeling study. *Pflügers Archiv-European Journal of Physiology*, 468(10):1651–1661, 2016.

[13] Chia-Chu Chiang, Rajat S Shivacharan, Xile Wei, Luis E Gonzalez-Reyes, and Dominique M Durand. Slow periodic activity in the longitudinal hippocampal slice can self-propagate non-synaptically by a mechanism consistent with ephaptic coupling. *The Journal of physiology*, 597(1):249–269, 2019.

[14] RK Nagle, EB Saff, and AD Snider. *Fundamentals of Differential Equations and Boundary Value Problems*. Pearson, 7th edition, 2017.

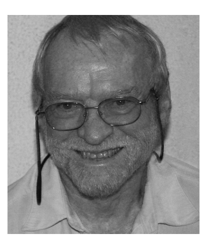
[15] Alwyn C. Scott. *Neuroscience: A Mathematical Primer*. Springer-Verlag, Inc., New York, 2002.

[16] Anna Corradi, Laura Croci, Vania Broccoli, Silvia Zecchini, Stefano Previtali, Wolfgang Wurst, Stefano Amadio, Roberto Maggi, Angelo Quattrini, and G Giacomo Consalez. Hypogonadotropic hypogonadism and peripheral neuropathy in ebf2-null mice. *Development*, 130(2):401–410, 2003.

[17] Leroy T Brown. Projections and termination of the corticospinal tract in rodents. *Experimental Brain Research*, 13(4):432–450, 1971.

Aman Chawla holds a B.S. in ECE from Cornell University and an M.S. in EECS from the Massachusetts Institute of Technology. He received a doctorate in Electrical Engineering from the University of South Florida in 2017, following which he has held two postdoctoral positions and is currently an Associate Professor at Alliance University in India. His research interests are at the interface of Electrical Engineering and Neuroscience.

Dr. Sal Morgera has focused on networks his entire professional life. He, and the teams he has directed, are responsible for the Canadian oceanographic data gathering networks, the acoustic networks for American submarine tactical and strategic communications, the worldwide CAT3 auto-landing networks for commercial aircraft, and the military wireless networks used in sensitive and challenging parts of the world. In the last decade, he has turned his attention to the neurological networks of the brain and their electric near-field and bio-metamaterial structures. Dr. Sal Morgera has published 276 fully refereed papers in journals and conference proceedings, 3 books, 2 of which are research monographs; and, recently, 2 articles in The Conversation. He is a Life Fellow of the Institution for Electrical and Electronics Engineers (IEEE), Fellow of the American Association for the Advancement of Science (AAAS), and Fellow of the Institution for Engineering and Technology (IET) for his many years in University Administration; development of a graduate Bioengineering Degree Program, which has become a model for the nation; and advances in our understanding of brain structure and function. Dr. Sal Morgera is Professor of Electrical Engineering, Professor of Bioengineering, and Tau Beta Pi Eminent Engineer at the University of South Florida, Emeritus Professor at Universities in Canada and the US, and Director of the Global Center for Neurological Networks. He received his Bachelor, Master, and Ph.D. degrees from Brown University, Providence, RI.



Arthur David Snider holds degrees from M.I.T. (B.S. mathematics, 1962), Boston U. (M.A. physics, 1966), and N.Y.U. (Ph.D. mathematics, 1971). He is retired from the faculties of electrical engineering, mathematics, and physics at the University of South Florida. Previous employment includes 4 years with the Draper Lab at M.I.T. He has coauthored 7 textbooks in applied mathematics and 90 research articles. Current research areas include neuron and semiconductor modeling.